IR Proximity Power Budget

Team Number: 104
Project Name: **EGR 304 Haptic Gripper**
Team Member Names: **Joseph, Sakiya, Jared, Emmanuel, Aakash**
Version: **1**

All Major Components	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Curiosity Nano	PIC18F57Q43	+1.8-5.5V	1	350	350	mA
	Ir Emitter	TSAL6400	+12 - 24V	1	200	200	mA
	IR Receiver	TSSP4056	+2.5 - 5.5V	1	5	5	mA
	Opamp	MCP6004	+5 to -5	1	100	100	mA
	5V regulator	LM7805	+5V - 35V	1	8	8	mA
	Wifi transceiver	(full part number)	+1.8 - 3.3V	1	350	350	mA

+5V Power Rail	Component Name	Part Number	Supply	Qty.	Absolute	Total	Unit
	Curiosity Nano	PIC18F57Q43	+1.8-5.5V	1	350	350	mA
	Ir Emitter	TSAL6400	+12 - 24V	1	200	200	mA
	IR Receiver	TSSP4056	+2.5 - 5.5V	1	5	5	mA
	Opamp	MCP6004	+5 to -5	1	100	100	mA
	5V regulator	LM7805	+5V - 35V	1	8	8	mA
	Wifi transceiver	(full part number)	+1.8 - 3.3V	1	350	350	mA
					Subtotal	1013	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	1266.25	mA
Regulator	+5V Regulator	LM7805	(range)	1	1500	1500	mA
					Total Remaining Current Available on +5V Rail	233.75	mA

External Power Source 1	Component Name	Part Number	Supply	Output	Absolute	Total	Unit
Power Source 1 Selection	Plug-in Wall Supply	(full part number)	110VAC	+24V	5000	5000	mA
Power Rails Connected to External Power Source 1	+5V Regulator	LM7805	(range)	1	1500	1500	mA
					Total Remaining Current Available on External Power Source 1	3500	mA